Team Details

- ➔ Team name - **SuperPumped**
- ➔ Team leader name - **Kush**
- ➔ Problem Statement - **Residential societies lack a proactive, privacy-preserving way to automatically detect unauthorized parking and stray animal threats from 24/7 CCTV feeds without relying on expensive, privacy-invasive cloud AI.**

Team SuperPumped

ColonyEdge: Privacy-First Edge AI

Upgrading residential security with AMD Ryzen AI NPUs for local, proactive threat detection and data privacy.

Kush,Luv
Enter the person's role or job title.

Market Gap

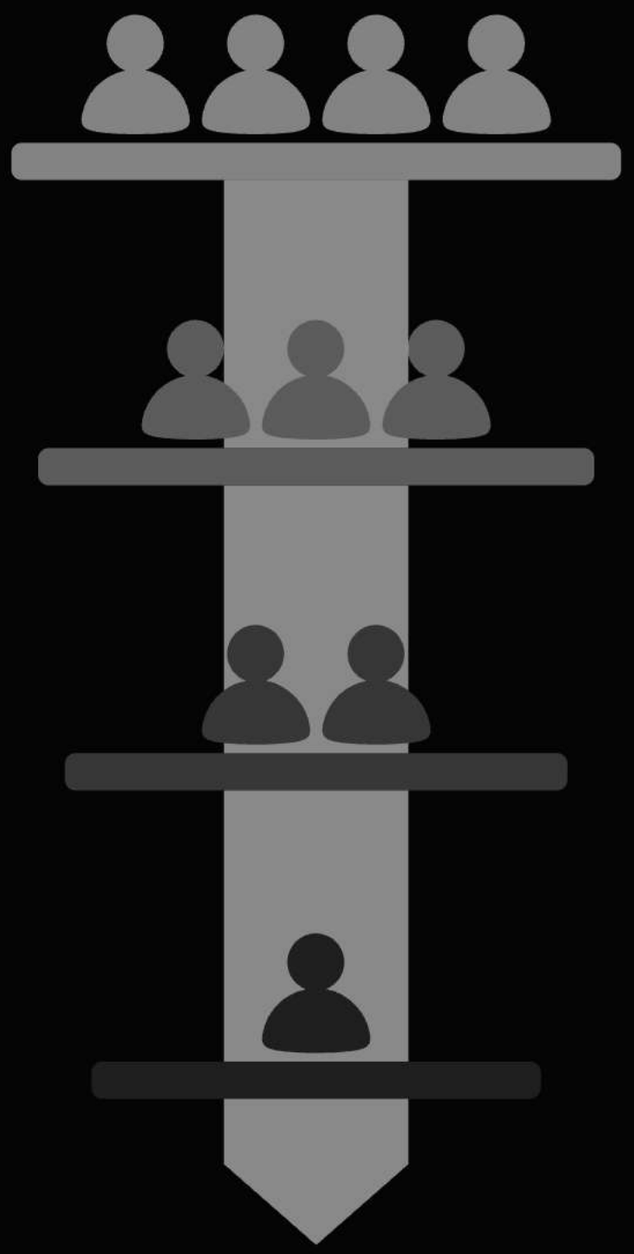
The Edge AI Opportunity

Moving from high-latency, privacy-invasive cloud processing to secure 100% offline edge inference.

Feature	Traditional Cloud AI	ColonyEdge (Edge AI)
Data Privacy	Video streamed to 3rd party servers	100% Local. No raw video leaves premises
Latency	High (Cloud round-trip)	Real-time (Inference at the source)
Bandwidth Costs	Crippling 24/7 streaming fees	Zero (LAN-based transmission)
Active Monitoring	Often reactive after incidents	Proactive AI-triggered alerts

Core Intelligent Features

Specialized computer vision modules designed for residential safety and vehicle management.



Stray Animal Detection

Real-time identification of dog packs in restricted zones like basements and play areas.

Smart Parking ALPR

Detects unauthorized vehicles by cross-referencing against a local flat-allocation database.

Live Command Center

React.js dark-mode dashboard providing instant WebSocket alerts for security guards.

Privacy-by-Design

Data is processed in milliseconds on the NPU and immediately discarded from RAM.

The Inference Pipeline

A streamlined workflow from raw video capture to actionable security alerts.

1

Capture

Society CCTV cameras capture live video frames and send them to the edge node.

2

Inference

Local AMD NPU runs YOLOv8 model (Class 16 for Dogs, Class 2 for Cars).

3

Logic Gate

System checks the cooldown timer and verifies the local parking database.

4

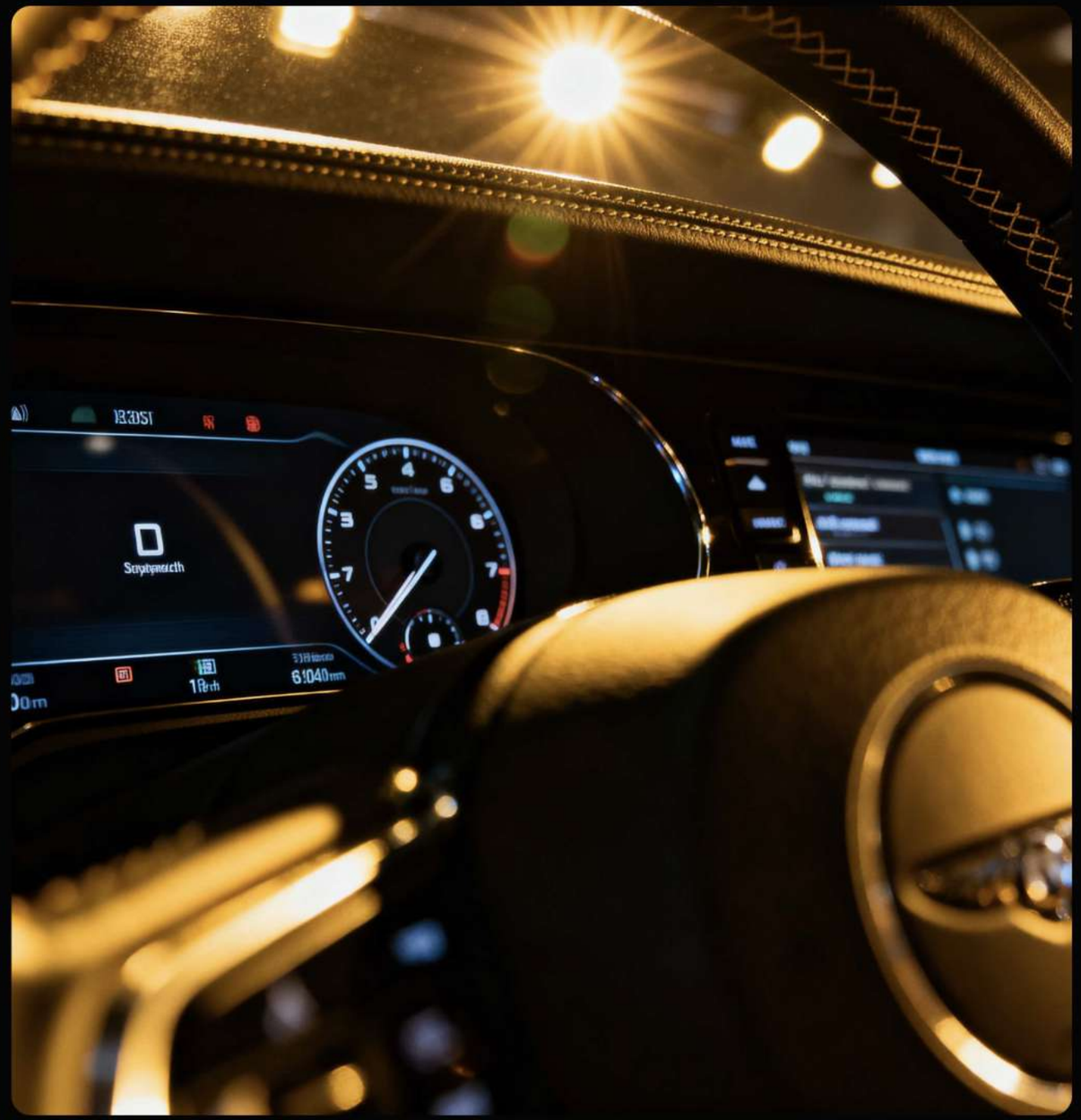
Transmission

Edge node fires a lightweight JSON payload via high-speed WebSockets.

5

Response

React dashboard flashes color-coded alerts (Red for Threat, Yellow for Warning).



User Interface

Security Command Center

Intuitive visual interface designed for immediate incident response by security personnel. The ColonyEdge dashboard is built with React.js and Tailwind CSS, featuring a high-contrast dark mode for 24/7 readability. It receives instant WebSocket payloads from the edge node, triggering state-based Red/Yellow alerts. Security personnel see a snapshot of the detection event along with precise timestamps and location data, allowing for rapid deployment of on-site guards.

System Architecture

A robust full-stack architecture optimized for high-performance edge computing.

Hardware Edge

Standard CCTV feeds integrated with local Security PC powered by AMD Ryzen AI NPU.

Software Edge

OpenCV frame extraction, YOLOv8 detection, and FastAPI WebSocket server.

Frontend Interface

Local Area Network (LAN) delivery to a stateful React.js Security Dashboard.

Technology

The Tech Stack

Leveraging industry-standard tools optimized for decentralized AI deployment.

Computer Vision & AI

Ultralytics YOLOv8 Nano optimized for edge OpenCV (cv2) for frame extraction Python 3 for core logic integration

Backend Edge Server

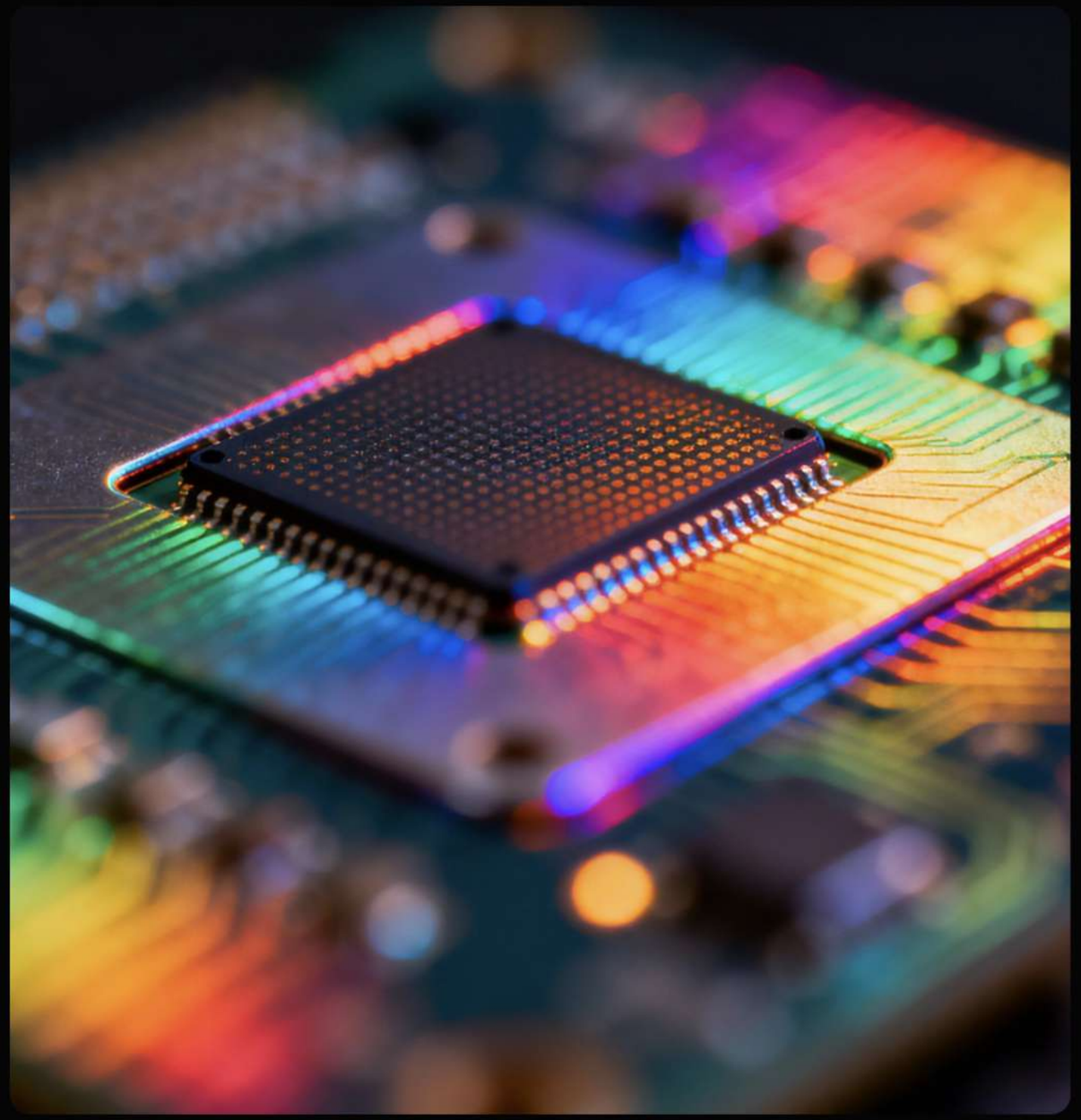
FastAPI for high-performance routing Asyncio & Uvicorn for non-blocking I/O WebSockets for zero-latency alerts

Frontend Interface

React.js for dynamic UI state Tailwind CSS for responsive styling State-based alerts for threat monitoring

AMD Integration

AMD Ryzen AI NPU Advantage



Unlocking 24/7 computer vision through specialized hardware efficiency. Continuous 24/7 inference on standard CPUs leads to thermal throttling and high power costs. ColonyEdge offloads matrix multiplications to the AMD Ryzen AI NPU, enabling real-time processing at a fraction of a watt. This hardware efficiency makes deploying decentralized, local AI financially viable for residential societies, directly utilizing AMD's push for intelligent edge computing.

Cost Analysis

Implementation & ROI

Comparing the long-term financial sustainability of Edge AI vs. Cloud solutions.

Category	Cloud-Based Security	ColonyEdge (AMD Node)
Monthly Fees	$500+ (Streaming & Compute)	$0 (No Recurring Cost)
Software Cost	Proprietary Subscriptions	Open-Source Framework
Hardware	Basic Cameras	One-time AMD Mini-PC purchase
Long-term ROI	Low (Rising Cloud Costs)	High (Pays for itself in months)

Next Steps

Prototype & Future Vision

Access the open-source repository and view the system in action.

AMD
Slingshot
HUMAN IMAGINATION
BUILT WITH AI
Powered by H2S
Thank you!